**Deployment of PrestaShop on AWS EC2 Instance**

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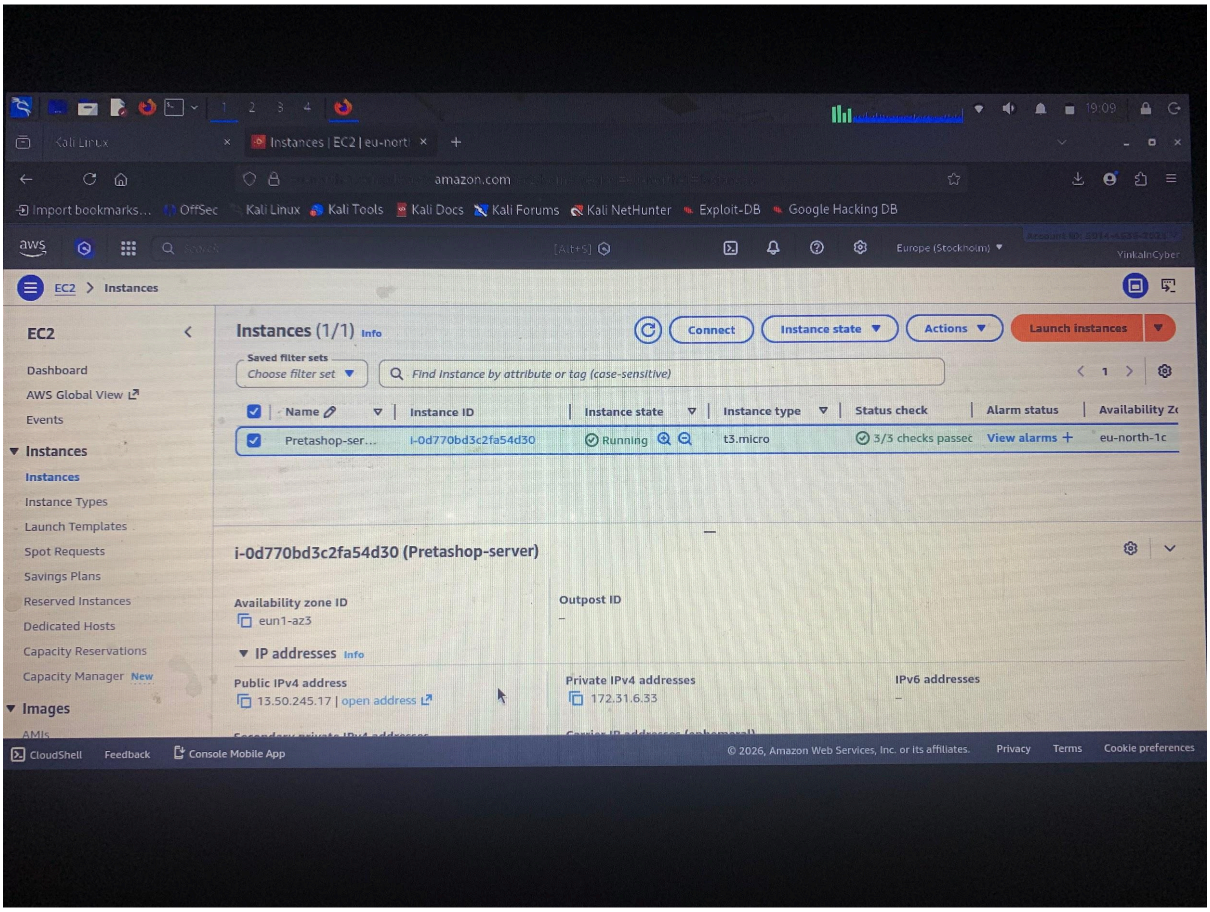
1. Introduction

This project demonstrates the deployment of an open-source e-commerce application, PrestaShop, on an AWS cloud infrastructure using an EC2 Ubuntu instance. The objective is to create a publicly accessible web server using AWS Free Tier services while applying basic cloud security and networking configurations.

2. AWS EC2 Instance Setup

An EC2 instance was launched using the AWS Free Tier with the following configuration:

- Instance Name: Prestashop-server
- Operating System: Ubuntu Server (Latest LTS)
- Instance Type: t2.micro (Free Tier eligible)
- Key Pair: SSH key pair generated during instance creation
- Public IP Address: Assigned automatically by AWS




# 3. Security Group Configuration

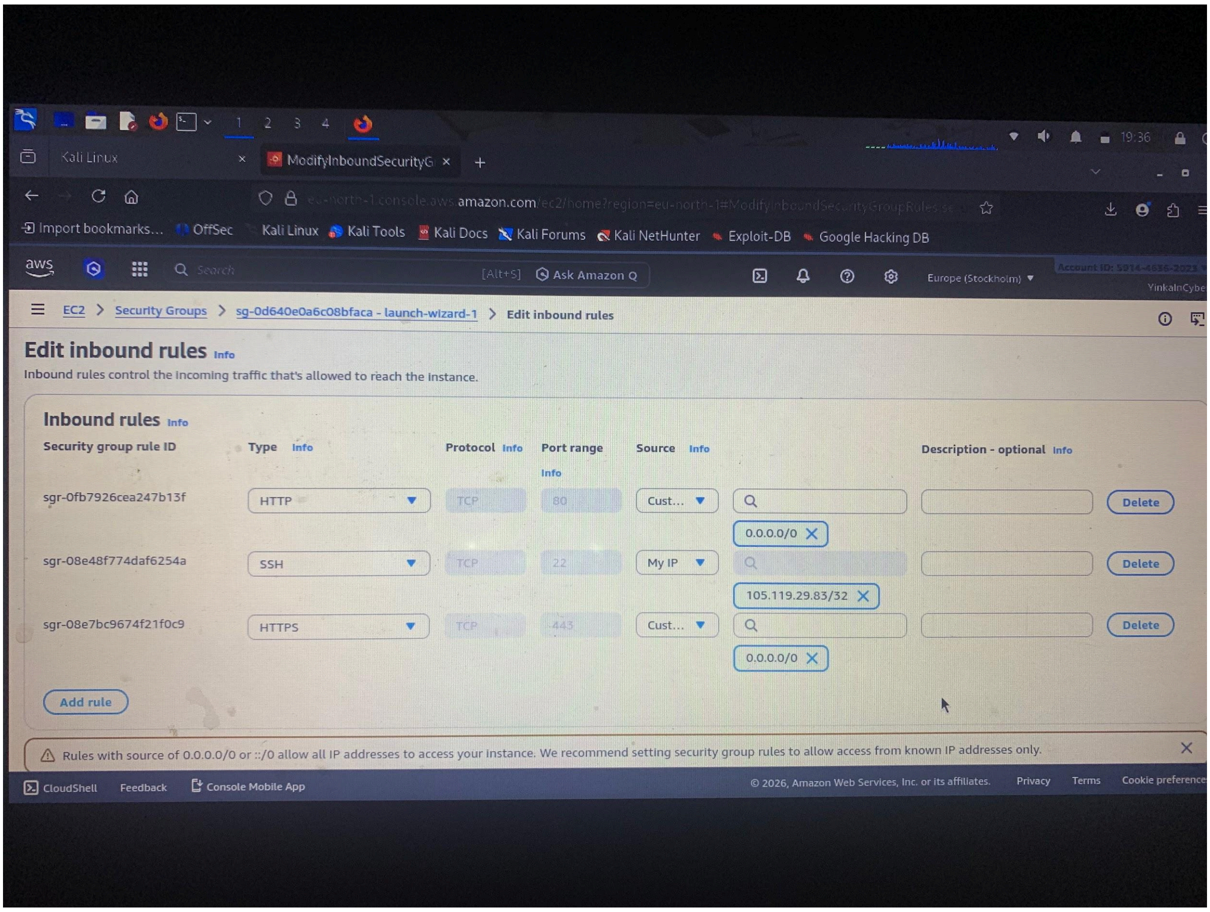
The security group was configured to allow controlled access to the EC2 instance.

Inbound rules include:

- SSH (22): restricted to “My IP” for secure administrative access
- HTTP (80): allowed from 0.0.0.0/0 for public web access
- HTTPS (443): allowed from 0.0.0.0/0 for secure web access

| TYPE | PORT | Source |
|---|---|---|
| SSH | 22 | My IP (initial) / 0.0.0.0/0 (temporary troubleshooting) |
| HTTP | 80 | 0.0.0.0/0 |

| | | |
|---|---|---|
| HTTPS | 443 | 0.0.0.0/0 |




# 4. Network Configuration (VPC, Subnet & Routing)

The EC2 instance was launched inside the default AWS Virtual Private Cloud (VPC), which provides networking capabilities for cloud resources.

A public subnet was used to ensure the instance could be accessed over the internet. Internet connectivity was enabled through an Internet Gateway attached to the VPC.

A route table was configured and associated with the subnet to allow outbound and inbound internet traffic.

## Route Table Configuration:

- Destination: 0.0.0.0/0
- Target: Internet Gateway (IGW)

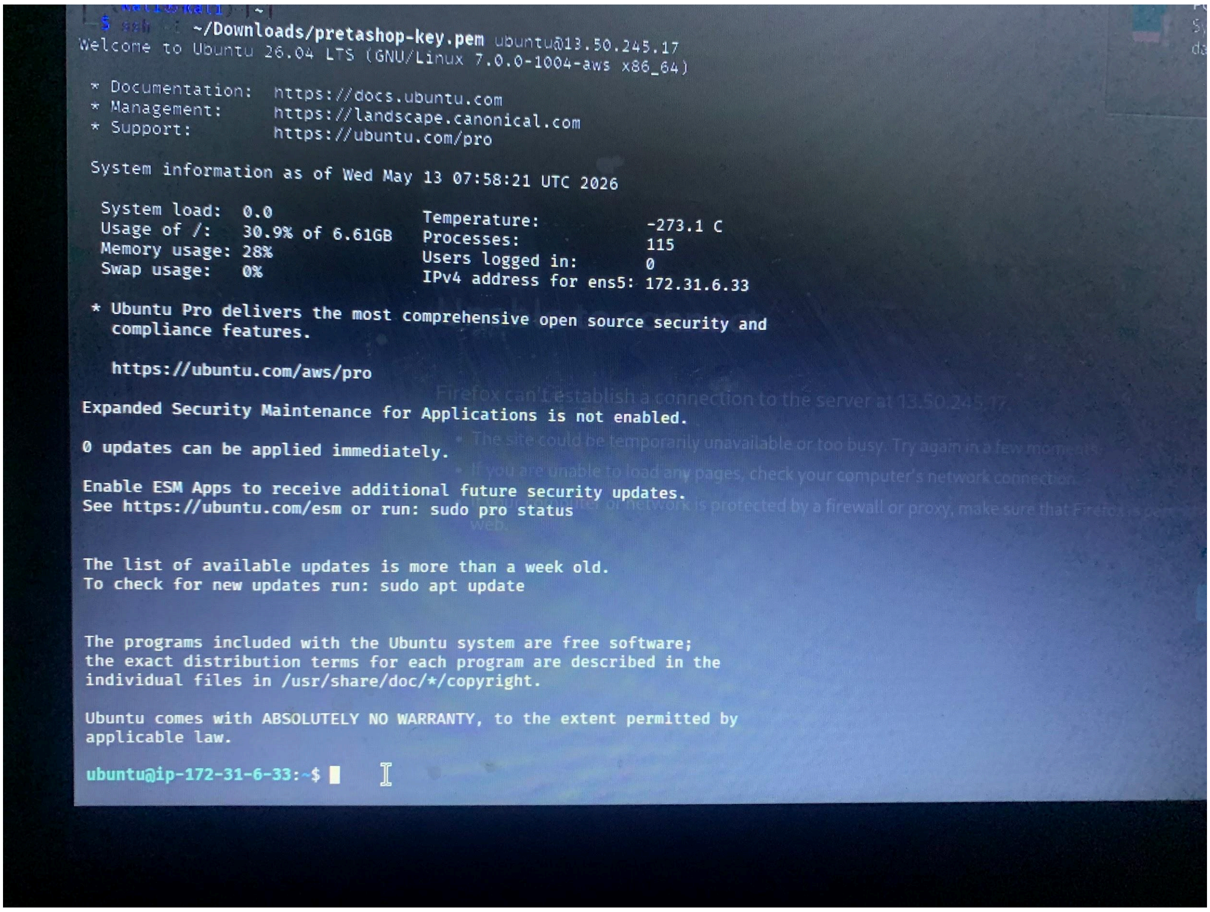
This configuration allows the EC2 instance to communicate with the internet and be accessible via its public IPv4 address.

| Destination | Target | Status |
|---|---|---|
| 0.0.0.0/0 | igw-033ff2… | Activ |
| 172.31.0.0/16 | local | Activ |

# 5. Server Access (SSH Connection)

The EC2 instance was accessed remotely using SSH from a local machine (Kali Linux environment). With this command: ssh -i ~/Downloads/prestashop-key.pem ubuntu@<Public-IP>

Successful authentication confirmed secure remote access to the server.




# 6. Apache Web Server Installation

Apache was installed to serve web content on the EC2 instance.

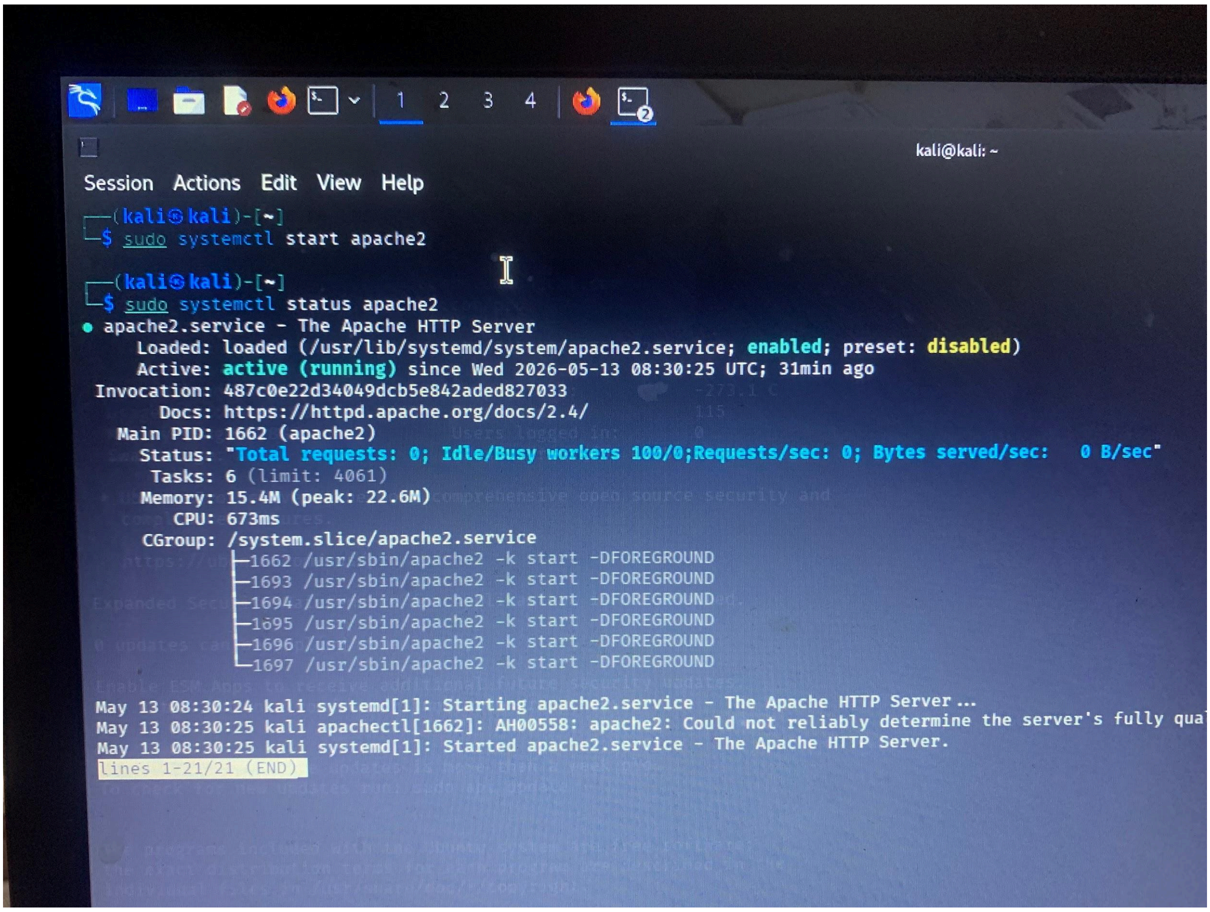
**Installation Commands: sudo apt update**

**sudo apt install apache2 -y**

**Service Status Check:**

sudo systemctl status apache2

Apache was confirmed running successfully.




# 7. Web Server Testing and Accessibility

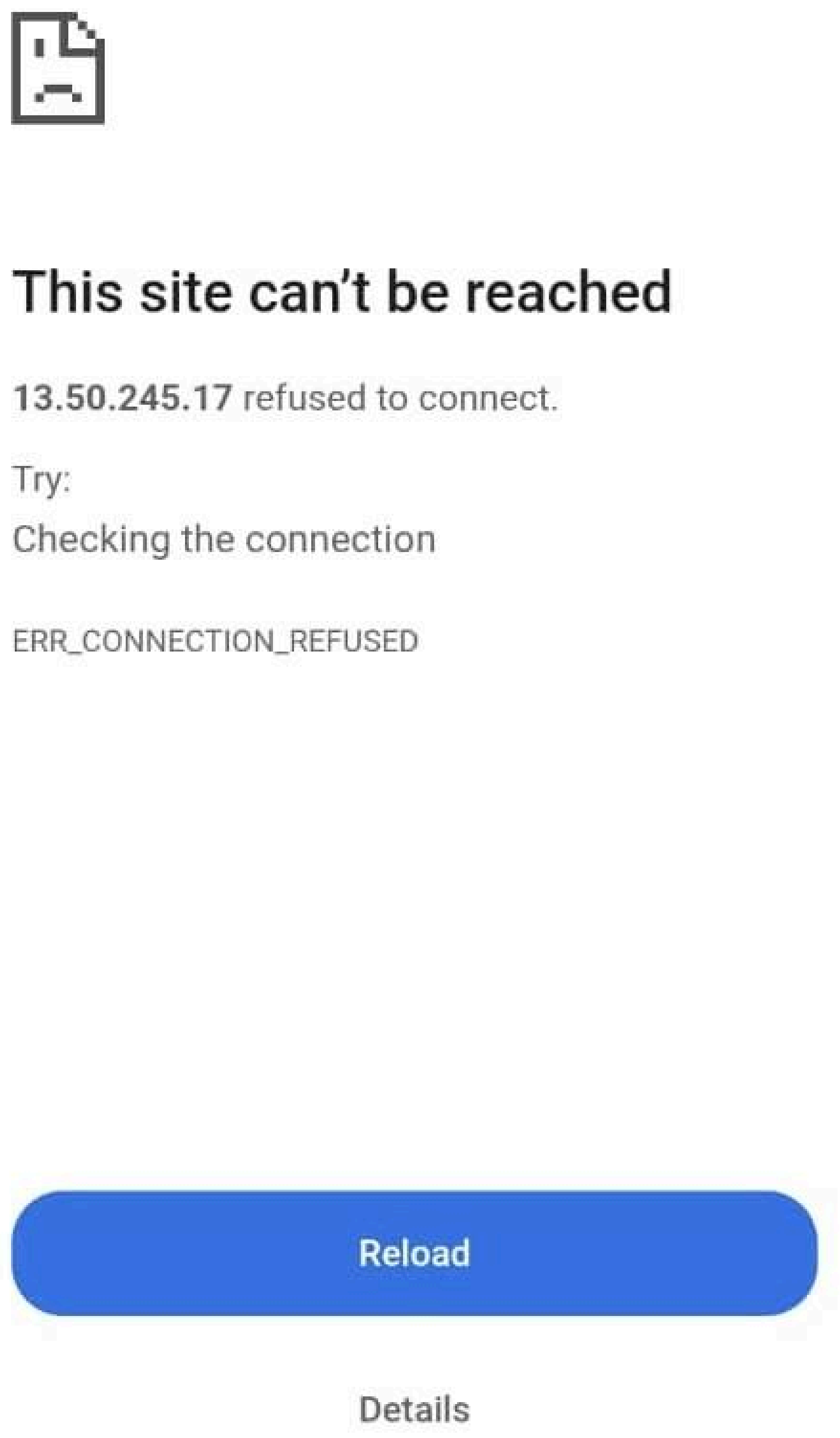
The Apache web server was tested using the EC2 instance public IPv4 address:http://13.50.245.17

Initial internal testing confirmed that the Apache service is running successfully on the EC2 instance using local access (localhost), indicating that the web server is properly installed and active.

However, external browser access from the internet was not fully successful during the testing phase. The following troubleshooting steps were performed:

- Verified EC2 instance status (running state confirmed)
- Confirmed security group configuration allowing HTTP (port 80)
- Confirmed route table includes Internet Gateway (IGW)
- Verified Apache service is active and running on the instance
- Established successful SSH connection to the instance

Despite these configurations, browser access to the public IP was not completed successfully within the testing timeframe.



# 8. PrestaShop Deployment (Preparation Stage)

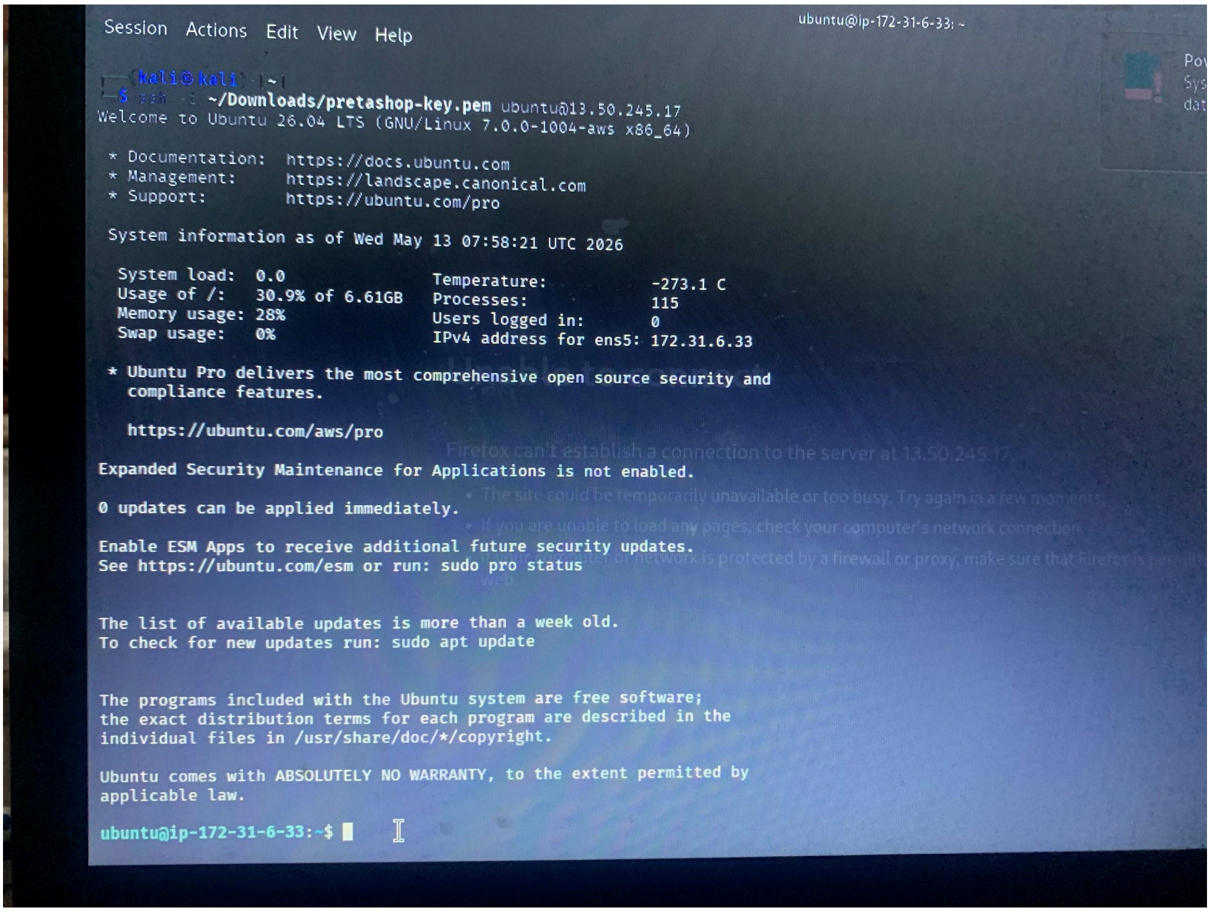
PrestaShop Official Website was identified as the target application for deployment on the EC2 instance.

The server environment was prepared by successfully installing and configuring Apache on the Ubuntu EC2 instance. This provided the required web server foundation for hosting the application.

However, the full PrestaShop installation was not completed within the scope and timeframe of this assessment.

The following components were identified as the next deployment steps:

- Installation of required PHP dependencies
- Configuration of a separate database using AWS RDS (MySQL)
- Completion of the PrestaShop web-based installation wizard




# 9. Database Architecture Requirement

In line with cloud architecture best practices, the database layer is designed to be hosted separately from the application server.

For this deployment, AWS RDS (MySQL) was identified as the preferred database solution instead of installing the database directly on the EC2 instance.

This approach follows a multi-tier architecture model consisting of:

- Presentation Layer: Web browser access
- Application Layer: EC2 Ubuntu server running Apache
- Data Layer: AWS RDS (MySQL database service)

### Benefits of this architecture include:

- Security: Separation of database from public-facing server reduces exposure risk
- Scalability: Database resources can be scaled independently of the application server
- Reliability: Managed database service improves uptime and reduces maintenance overhead
- Best Practice Compliance: Aligns with standard cloud deployment architecture principles

**Note:** No RDS instance was deployed during this phase; this section describes the intended architecture design.

# 11. Explanation of External Access Failure

During testing, external browser access to the Apache web server using the EC2 public IP address was not successful. Several potential causes were identified through troubleshooting analysis.

Possible Causes:

- Apache Binding to Localhost:

  Apache may be configured to listen only on 127.0.0.1 instead of all interfaces (0.0.0.0), preventing external access.

- Security Group Misconfiguration:

  Although HTTP port 80 was opened, incorrect or temporary firewall settings could restrict inbound traffic.

- Network ACL Restrictions:

  AWS Network Access Control Lists (ACLs) may block inbound or outbound traffic if not properly configured.

- Missing or Changing Public IP (Elastic IP not used):

  The instance uses an auto-assigned public IPv4 address, which may change after restart, affecting accessibility.

- UFW (Linux Firewall) Configuration:

  Ubuntu firewall settings may block incoming HTTP traffic if enabled.

- Browser or DNS Caching Issues:

  Local browser caching or ISP-level DNS resolution delays may affect connectivity testing.

### 12. Cybersecurity Hardening Measures

To improve the security posture of the deployed EC2 environment, several best practices were considered:

- SSH Key Authentication:

  Secure login was implemented using SSH key pairs instead of password authentication.

- Disabling Root Login (Recommended):

  Direct root login should be disabled to reduce attack surface.

- System Updates:

  Regular updates should be applied using:sudo apt update && sudo apt upgrade -y

# 13. Conclusion

This project demonstrates the deployment of a cloud-based web server using Amazon Web Services (AWS). An Ubuntu EC2 instance was successfully created and configured to host a web server environment.

Key implementations in this project include:

- Creation and configuration of an AWS EC2 instance
- Setup of security groups to manage inbound traffic

- Configuration of VPC routing for internet connectivity
- Installation and activation of the Apache web server
- Verification of internal server functionality through SSH access

Although external browser access to the web server was not fully successful during testing, the core infrastructure setup and server configuration were completed and validated.

The project also highlights best practices in cloud architecture, including separation of application and database layers using AWS RDS and implementation of basic cybersecurity hardening principles.

Overall, this work demonstrates practical understanding of cloud computing, Linux server management, and AWS infrastructure deployment.

# References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